PRODUCT VISION & STRATEGY DOCUMENT

RESORA

Complete Product Vision, Feature Architecture, Production Roadmap & Commercial Strategy

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SECTION 01

Executive Product Vision

The current Resora project already has a strong code-level baseline, but not everything should be considered production-ready. Some features may already be working, some may be partially implemented, some may still use mock data or placeholder logic, and some may require real external services, stronger error handling, better scalability, or additional testing.

More importantly, we should change the way we think about Resora.

We should not think:

> *“This is our FYP, so we need a frontend, backend, AI, and database.”*

Instead, we should think:

> *“We are building the first production version of a real research technology product.”*

If we want to deploy Resora after the FYP, market it, acquire real researchers as users, introduce subscriptions, continue improving it, and potentially turn it into a long-term startup, then the architecture and product decisions we make now should support that future.

Therefore, we should not divide the project only into:

- Frontend
- Backend
- AI
- Database

Instead, we should divide the product into complete features and modules. For example, one team member can own the complete “Research Search” module, including:

- Frontend
- Backend
- APIs
- Database
- Search logic
- Ranking
- Deduplication
- Error handling
- Testing
- Documentation

Another member can own “Paper Intelligence,” including:

- PDF processing
- OCR
- Extraction
- Summarization
- RAG
- Evidence
- AI interaction

This gives every feature clear ownership.

The long-term vision

> *Resora is an AI-powered Research OS that helps researchers discover, understand, verify, compare, organize, analyze, write, and continue research from one platform.*

The overall research workflow should become:

Discover → Search → Understand → Read → Ask → Compare → Verify → Organize → Analyze → Find Gaps → Write → Collaborate → Monitor → Continue Research

The goal is not to become just another paper search engine. The goal is to become a researcher’s complete working environment.

SECTION 02

What Resora Should Ultimately Solve

A researcher currently has to use many different tools:

Tool	Used for
Google Scholar / Semantic Scholar	Finding papers
ResearchRabbit / Connected Papers	Discovering relationships
PDF readers	Reading papers

Tool	Used for
ChatGPT / Claude / other AI tools	Asking questions
Elicit / Consensus / other research AI tools	Literature analysis
Zotero	Citations
Notion / Obsidian / Google Docs	Notes
Excel / spreadsheets	Paper comparison
GitHub	Code
Google Drive	Documents
Email / Slack / Discord	Collaboration
Manual spreadsheets	Literature review screening

This creates a fragmented workflow. Resora should gradually bring these activities together. The researcher should not have to constantly move between ten different tools. The long-term idea is:

> *One research environment instead of many disconnected research tools.*

SECTION 03

Current Resora Baseline

The current project already gives us a good starting point. The existing baseline includes or is designed around the following capabilities.

3.1 Multi-Source Research Paper Search

Resora can search research literature from sources such as:

- OpenAlex
- arXiv
- Crossref
- Semantic Scholar
- Other academic APIs and databases that can be added later

The important thing is that users should not have to search every source individually. Resora should aggregate and normalize results.

SECTION 04

Smart Search, Filters & Deduplication

The search system should support:

- Keyword search
- Natural-language search
- Semantic search
- Hybrid search
- Boolean operators
- DOI search
- Title search
- Topic search
- Citation search
- Similar paper search

- Exact phrase search
- Author search
- Citing paper search
- Reference search

Filters

- Publication year
- Author
- Institution
- Journal
- Conference
- Citation count
- Topic
- Method
- Model
- Dataset
- Task
- Open-access status
- Peer-reviewed / preprint status
- Publication type

Deduplication

If the same paper appears from multiple sources, Resora should automatically detect and merge duplicates. The user should see one clean research record rather than five copies of the same paper.

SECTION 05

AI Research Search

Normal search is only the beginning. A user should be able to ask:

> *“What are the latest approaches for reducing hallucinations in multimodal LLMs?”*

Resora should understand the research question and perform a research-oriented search. The system should:

1. Understand the question.
2. Identify important concepts.
3. Generate search queries.
4. Search multiple sources.
5. Collect candidate papers.
6. Remove duplicates.
7. Rank results.
8. Group papers by approach.
9. Extract important findings.
10. Identify supporting evidence.
11. Identify conflicting findings.
12. Identify emerging approaches.
13. Identify potential research gaps.
14. Present the relevant papers.
15. Explain why those papers were selected.

This should feel like:

> *“Research with Resora”*

rather than simply:

> *“Search papers.”*

SECTION 06

Research Database

One of the most important long-term decisions is to build our own normalized research database. We should not depend completely on external APIs every time a user searches. External sources should feed our system.

Our internal database should gradually contain structured information about:

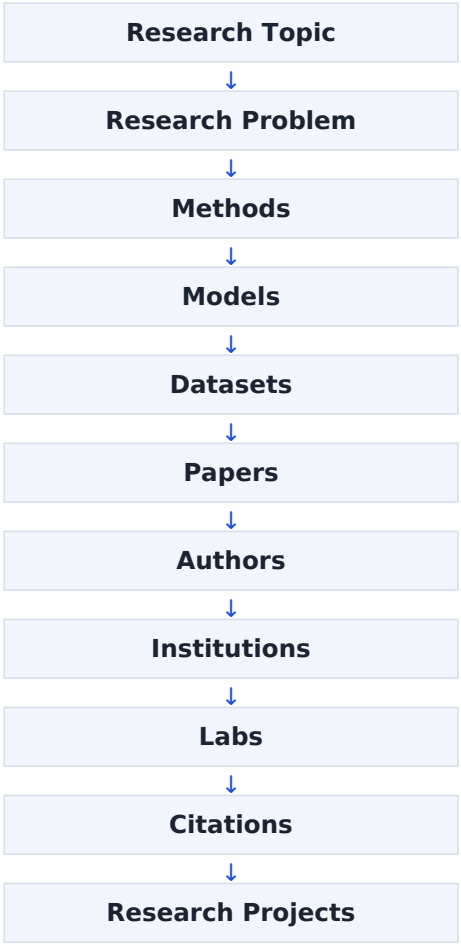
- Papers
- Authors
- Institutions
- Topics
- Methods
- Models
- Datasets
- Tasks
- Metrics
- Results
- Citations
- References
- Journals
- Conferences
- Research projects
- Code repositories
- Research relationships

This database becomes one of Resora’s long-term assets.

SECTION 07

Research Knowledge Graph

The database should eventually evolve into a Research Knowledge Graph. For example:



A paper should not exist as an isolated document. Resora should understand relationships between research entities. For example:

EXAMPLE RELATIONSHIPS

```
Paper A
├── cites          → Paper B
├── uses           → Dataset X
├── proposes       → Method Y
├── uses           → Model Z
├── written by     → Author A
└── implemented in → GitHub Repo
```

This can later power:

- Recommendations
- Research trends
- Gap detection
- Similarity
- Researcher discovery
- Institution discovery
- Dataset discovery
- Citation intelligence
- Research-agent workflows

SECTION 08

AI Paper Reader

When a researcher opens a paper, Resora should not just display the PDF. It should understand the paper. It should extract:

- Title
- Abstract
- Research problem
- Research questions
- Motivation
- Contributions
- Methodology
- Dataset
- Sample size
- Models
- Baselines
- Experiments
- Metrics
- Results
- Limitations
- Future work
- References
- Figures
- Tables
- Equations
- Code availability
- Dataset availability

The researcher should be able to ask:

“What problem does this paper solve?”
“What is the main contribution?”
“Why did they choose this method?”
“What dataset did they use?”
“What are the main results?”
“What is the strongest experiment?”
“What are the limitations?”
“What future work do the authors suggest?”
“What could I improve?”

The answers should be grounded in the actual paper.

SECTION 09

Advanced PDF Intelligence

PDF processing should become much stronger than basic text extraction. The system should eventually understand:

- Text
- Headings
- Tables
- Figures
- Captions
- Equations
- References
- Footnotes
- Supplementary material

It should preserve relationships. For example:

> *“What does Figure 3 show?”*

Resora should understand Figure 3 and its caption. Similarly:

> *“What are the results in Table 2?”*

should return the relevant table information. This requires proper document parsing and potentially OCR for scanned PDFs.

SECTION 10

Evidence-Grounded AI

This should be one of the core principles of Resora. AI should not simply produce an answer and say:

> *“According to the paper...”*

Instead, Resora should show:

- Source paper
- Page
- Section
- Supporting passage
- Evidence
- Citation
- Confidence / evidence strength where appropriate

The user should be able to click the evidence and verify it. This creates trust. The goal should be:

> *AI answer + evidence + source + verification*

rather than:

> *AI answer only*

SECTION 11

Multi-Paper Intelligence

This should be one of Resora’s flagship features. A user can select:

- 2 papers
- 5 papers
- 10 papers
- 15 papers
- Eventually larger collections where infrastructure allows

Then click “Compare.” Resora creates a structured table. Example:

Paper	Problem	Method	Dataset	Model	Metric	Result	Limitation
Paper A	...	...	...	...	...	...	...
Paper B	...	...	...	...	...	...	...

But comparison should go beyond fixed columns. The user should be able to say:

“Add training dataset size.”
“Add computational cost.”
“Add whether code is available.”
“Add the main limitation.”

Resora should extract that information from all selected papers. This can save researchers hours of manual spreadsheet work.

SECTION 12

Research Gap Discovery

Research gap detection needs to be handled carefully. We should not simply tell researchers:

> *“This is your research gap.”*

AI cannot guarantee that a research gap truly does not exist. Instead, Resora should provide evidence-based gap signals. For example:

EXAMPLE GAP SIGNAL

```
Topic: Multimodal LLM Hallucination

82 papers identified

63 use Benchmark A
11 investigate multilingual settings
4 investigate low-resource languages
2 directly compare Method X and Method Y
```

Then Resora can say:

> *Potentially under-explored area*

and show the papers supporting that observation. The system should distinguish:

- Strong evidence
- Moderate evidence
- Limited evidence
- AI-generated hypothesis

This is much safer and more useful for researchers.

SECTION 13

Contradiction & Evidence Engine

Suppose:

Paper A	Paper B
“Method X improves performance.”	“Method X does not significantly improve performance.”

Resora should detect the conflict. Then investigate:

- Dataset differences
- Experimental setups

- Population differences
- Model differences
- Evaluation metrics
- Sample sizes
- Benchmarks
- Publication years
- Research conditions

The result should include:

Section	Meaning
Supporting Evidence	Which papers support the finding?
Opposing Evidence	Which papers disagree?
Mixed Evidence	Which papers report conditional or mixed results?
Possible Reasons	What methodological differences may explain the disagreement?

This is more valuable than simply displaying “70% of papers agree.”

SECTION 14

Research Consensus View

A dedicated research evidence view could summarize a topic. For example:

EXAMPLE

```
Research Question: Does Method X improve performance?

Supporting studies : 14
Opposing studies   : 5
Mixed findings     : 4
```

Then show:

- Papers
- Evidence
- Study characteristics
- Dataset differences
- Method differences
- Limitations

This gives researchers a transparent view instead of hiding the complexity behind one score.

SECTION 15

Personalized Research Recommendations

Resora should understand the user’s research interests. Suppose the user frequently researches:

- RAG
- Multimodal LLMs
- Hallucination
- Evaluation
- Small datasets

Resora should recommend papers based on the user’s actual research context. Each recommendation should explain, for example:

> *Recommended because it uses the same dataset.*

> *Recommended because it extends a method from one of your saved papers.*

> *Recommended because it addresses a limitation you previously identified.*

This makes recommendations genuinely useful.

SECTION 16

Research Alerts

Users should be able to follow:

- Topics
- Researchers
- Methods
- Datasets
- Journals
- Conferences
- Projects
- Research areas

They should receive:

- Daily alerts
- Weekly alerts
- Monthly alerts

EXAMPLE

```
Your Weekly Research Brief

- 18 new papers
- 5 highly relevant papers
- 2 emerging methods
- 1 new dataset
- 1 important researcher publication
- 2 papers related to your current project
```

This can become an important retention mechanism.

SECTION 17

Research Trends

For any research topic, Resora should show:

- Publication growth
- Emerging topics
- Popular methods
- Emerging methods
- Popular datasets
- Models
- Research clusters
- Researchers
- Institutions
- Citation trends
- Keyword trends

The user should be able to see:

> *“How has this field changed over the last 5–10 years?”*

This can be valuable for:

- Literature reviews
- Research proposals
- PhD students
- Researchers
- Labs
- Industry R&D teams

SECTION 18

Researcher Intelligence

Users should be able to search:

> *“Researchers working on multimodal hallucination.”*

Resora can provide:

- Publications
- Research topics
- Recent work
- Institutions
- Collaborators
- Research timeline
- Related researchers
- Research methods

Potential use cases include:

- Finding collaborators
- Finding supervisors
- Finding research labs
- RA discovery
- PhD research discovery

SECTION 19

Institution & Lab Intelligence

Users should be able to search:

> *“AI research at University X.”*

Resora can show:

- Researchers
- Labs
- Research groups
- Publications
- Research topics
- Recent work
- Research areas
- Collaboration patterns

Eventually, this can become a separate institutional product.

SECTION 20

Research Workspace

Every user should have a central workspace. Example:

EXAMPLE WORKSPACE

```
Project: Multimodal LLM Hallucination

Papers
Important Papers
Notes
Highlights
AI Analysis
Comparisons
Research Gaps
Literature Review
References
Researchers
Datasets
Code
Tasks
```

This should be the user's research home.

SECTION 21

Research Memory

Resora should remember research context. It can remember:

- Projects
- Research interests
- Saved papers
- Notes
- Searches
- Researchers
- Datasets
- Previous AI conversations

For example:

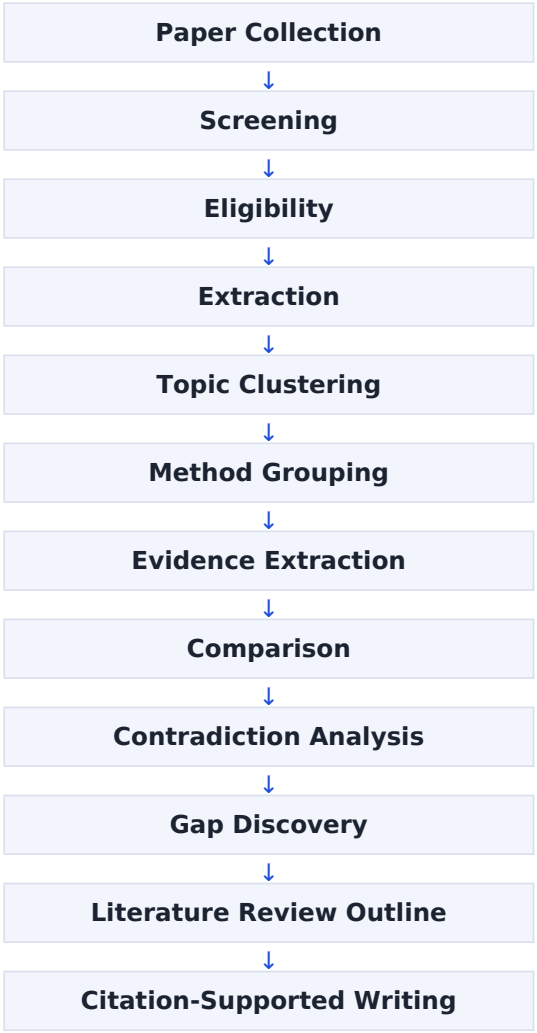
> *"Find papers related to the problem I was working on last semester."*

Resora should understand which project the user is talking about. This is where Resora becomes a Personal Research Brain rather than simply a search engine.

SECTION 22

Literature Review Builder

A user should be able to select 30–50 papers. Resora should support:



This can eventually support systematic and structured literature-review workflows.

SECTION 23

Screening & Review Management

This is an important area that should be added more seriously. Researchers conducting literature reviews often need to screen many papers. Resora should support:

- Include
- Exclude
- Maybe
- Reason for exclusion
- Reviewer notes
- Tags
- Duplicate detection
- Screening status
- Full-text availability
- Eligibility criteria
- Reviewer assignments

For teams, two researchers should eventually be able to independently screen papers and compare decisions. This makes Resora much more useful for real systematic-review workflows.

SECTION 24

PRISMA-Oriented Workflow

Since literature reviews often require structured screening, Resora could eventually support a PRISMA-style workflow. For example:



The system can maintain these counts automatically. This can save researchers a lot of manual work.

SECTION 25

AI Academic Writing

Resora should support:

- Literature review
- Introduction
- Related work
- Research proposal
- Research questions
- Abstract
- Conclusion
- Research outline

But the system should be research-grounded. For example:

> *“Write the related work section using only papers from this project.”*

Resora should generate the draft from the selected evidence. It should show citations and allow the user to inspect the sources.

SECTION 26

Citation Management

Users should be able to:

- Save citations
- Create folders
- Create collections
- Add tags
- Manage DOI
- Select citation styles
- Export BibTeX
- Export RIS
- Export EndNote
- Export CSL-JSON
- Export Markdown

Future integrations

- Zotero
- Microsoft Word
- LaTeX
- Overleaf

SECTION 27

Citation Intelligence

While users are writing, Resora should be able to check:

> *“Does this citation actually support this claim?”*

Example:

> *“Recent studies prove X.”*

Resora can respond:

> *This citation may not fully support the statement.*

Then suggest:

> *These papers provide stronger evidence for this claim.*

This can become an important differentiating feature.

SECTION 28

Claim Verification

A future writing mode could work like:

User writes a claim
↓
Resora identifies factual statement
↓
Searches selected research
↓
Checks evidence
↓
Suggests supporting papers
↓
Shows evidence strength

This is useful for reducing unsupported academic claims.

SECTION 29

Research Trust & Source Information

Every paper should have transparent source information:

- Peer-reviewed / preprint
- Journal / conference
- DOI
- DOI verification
- Open-access availability
- Publication date
- Citation information
- Retraction information
- Correction information
- Full-text availability

Possible warnings:

Warning label
Preprint
Retraction detected
Correction available
DOI unavailable
Full text unavailable

The objective should be transparency, not pretending that Resora can perfectly judge scientific quality.

SECTION 30

Dataset Intelligence

Resora should create relationships between:

> *Paper ↔ Dataset*

Users should be able to search:

> *“What datasets are commonly used for diabetes prediction?”*

Resora can show:

- Dataset
- Papers using it
- Tasks
- Models
- Metrics
- Sample size
- Access information
- License information

This can become very useful for researchers starting new projects.

SECTION 31

Code & Reproducibility Intelligence

For papers with public code:

> **Paper → GitHub → Dataset → Model**

Resora can show:

- Repository
- Programming language
- Framework
- Code availability
- Dataset availability
- Documentation
- Repository activity
- Implementation links

A future feature could be “Try this paper,” where users can access a reproducible environment if the infrastructure allows it.

SECTION 32

Reproducibility Checklist

Instead of claiming a universal scientific-quality score, Resora can provide a transparent checklist. Example:

Item	Status
Code available	✓
Dataset available	✓

Item	Status
Hyperparameters available	✓
Experimental details	✓
Pretrained model	✗
License	✓

Reproducibility items: 5/6. The criteria should always be visible.

SECTION 33

Research Project Management

Research is not only reading papers. Users should also manage:

- Tasks
- Deadlines
- Reading lists
- Questions
- Research ideas
- Experiments
- Notes
- Project milestones

For example:

EXAMPLE PROJECT OVERVIEW

```
Project: Multimodal Hallucination

To Read     : 17 papers
Important   : 8 papers
Compared    : 5 papers
Gaps Found  : 4
Research Ideas : 7
Tasks       : 12
```

This moves Resora toward a complete research workspace.

SECTION 34

Research Ideas

This is another feature we should add. When a user has a collection of papers, Resora can help turn that collection into possible research ideas.